Wireshark Installation Guide

Wireshark (`https://www.wireshark.org`) is a free, open-source packet analyser. In this lab you will use it to inspect the TCP messages exchanged between the fire trucks and the fire-department server — watching actual bytes on the wire is the most direct way to verify that your protocol works as designed.

Because the simulation runs entirely on your own machine (`localhost`), all traffic passes through the **loopback interface**. The name of this interface in Wireshark differs by operating system:

Operating system	Loopback interface name in Wireshark
Ubuntu Linux	`lo`
macOS	`lo0`
Windows	`Adapter for loopback traffic capture`

Follow the section for your operating system below.

1 Ubuntu Linux

1. Install Wireshark from the Ubuntu package repository:

```
sudo apt update
sudo apt install wireshark
```

   During installation you will be asked: *"Should non-superusers be able to capture packets?"* — select **Yes**.

2. Add your user account to the `wireshark` group so that you can capture packets without running Wireshark as root:

```
sudo usermod -aG wireshark $USER
```

3. **Log out and log back in** for the group membership to take effect. Alternatively, apply it immediately in the current terminal session:

```
newgrp wireshark
```

4. Start Wireshark:

```
wireshark &
```

   In the interface list, look for the entry named **lo** and double-click it to begin capturing loopback traffic.

2 macOS

1. Go to `https://www.wireshark.org/download.html` and download the installer that matches your Mac:

   - **Intel Mac:** select the *macOS Intel 64-bit .dmg*.

- **Apple Silicon (M1/M2/M3/M4):** select the *macOS Apple Silicon .dmg*.

2. Open the downloaded `.dmg` and drag **Wireshark** to your `Applications` folder.
3. **Grant capture permissions.** The Wireshark disk image also contains a helper package called **Install ChmodBPF**. Double-click it and follow the prompts. ChmodBPF adjusts the permissions of the BSD Packet Filter devices so that Wireshark can open network interfaces — including the loopback interface `lo0` — without requiring superuser rights. Without this step, Wireshark will start but will not be able to capture any traffic.

   > If you did not run ChmodBPF during installation you can install it later: open the Wireshark `.dmg` again and double-click **Install ChmodBPF**.

4. Start Wireshark from your Applications folder. If macOS blocks the app on first launch, go to **System Settings** → **Privacy & Security** and click **Open Anyway**.
5. In the Wireshark interface list, look for the entry named **lo0** and double-click it to begin capturing loopback traffic.

3 Windows

1. Go to `https://www.wireshark.org/download.html` and download the **Windows x64 Installer**.
2. Run the installer. When the installer reaches the **Npcap** step, make sure it is checked and click **Install**. Npcap is the packet-capture driver that Wireshark uses to access network interfaces on Windows, including the loopback adapter. Accept the Npcap licence and complete its installation before continuing with the Wireshark installer.

   > If you already have an older version of Npcap installed you may be prompted to upgrade it — do so to ensure loopback capture works correctly.

3. Start Wireshark from the Start menu.
4. In the interface list, look for the entry named **Adapter for loopback traffic capture** and double-click it to begin capturing loopback traffic. If this entry is not visible, verify that Npcap was installed successfully: go to **Settings -> Apps -> Installed apps**, search for "Npcap", and re-run the Wireshark installer if it is missing.

4 Verifying the installation

Whichever platform you are on, do a quick sanity check before the lab:

1. Start Wireshark and double-click the loopback interface (see table at the top of this guide for the name on your platform).
2. Open a terminal and run:

   ```
   curl http://localhost
   ```

   (It will fail with a connection error, but it will still generate TCP traffic on the loopback interface.)

3. In Wireshark you should see at least one TCP segment with destination port 80 appear in the packet list. If you do, capture permissions are set up correctly.

4. Stop the capture (red square button or **Capture** → **Stop**).

You are now ready for Task 2 of the lab.